# D Cheatsheets and Quick Reference

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## D.1 openalexR quick reference

```r
library(openalexR)

# Fetch works by journal (source)
works ← fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  type = "article"
)

# Fetch works by institution
works ← fetch_openalex(
  entity = "works",
  authorships.institutions.id = "I63966007",
  from_publication_date = "2022-01-01",
  type = "article"
)

# Fetch works by topic
works ← fetch_openalex(
  entity = "works",
  topics.id = "T10102",
  from_publication_date = "2023-01-01"
)

# Fetch author profile
author ← fetch_openalex(entity = "authors", id = "A5023888391")

# Fetch institution profile
inst ← fetch_openalex(entity = "institutions", id = "I63966007")

# Sampling
works ← fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  options = list(sample = 500, seed = 42)
)
```

## D.2 quanteda pipeline

```
library(quanteda)

# Full text-analysis pipeline
corp ← corpus(df, docid_field = "id", text_field = "text")
toks ← tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) ▷
  tokens_tolower() ▷
  tokens_remove(stopwords("en"))
dfmat ← dfm(toks) ▷
  dfm_trim(min_termfreq = 5, min_docfreq = 3)

# TF-IDF weighting
dfmat_tfidf ← dfm_tfidf(dfmat)

# Top features
topfeatures(dfmat, 20)

# Keyword-in-context
kwic(toks, pattern = "open access", window = 5)
```

## D.3 igraph / tidygraph / ggraph

```
library(igraph)
library(tidygraph)
library(ggraph)

# Build graph from edge list
g ← graph_from_data_frame(edges, directed = FALSE, vertices = nodes)

# Tidygraph conversion
tg ← as_tbl_graph(g) ▷
  activate(nodes) ▷
  mutate(
    degree = centrality_degree(),
    between = centrality_betweenness(),
    community = group_leiden(resolution = 1.0)
  )

# ggraph visualisation
ggraph(tg, layout = "fr") +
  geom_edge_link(aes(alpha = weight), show.legend = FALSE) +
  geom_node_point(aes(size = degree, colour = factor(community))) +
  geom_node_text(aes(label = name), repel = TRUE, size = 3) +
  theme_void()
```

## D.4 Common data wrangling patterns

```
library(tidyverse)

# Unnest OpenAlex authorships
authors ← works ▷
  select(work_id = id, authorships) ▷
  unnest(authorships, names_sep = "_")

# Unnest OpenAlex topics
topics ← works ▷
  select(work_id = id, topics) ▷
  unnest(topics, names_sep = "_")

# Compute h-index
compute_h_index ← function(citations) {
  citations ← sort(citations, decreasing = TRUE)
  sum(citations ⩾ seq_along(citations))
}

# Field normalisation
works ▷
  group_by(field, year) ▷
  mutate(
    field_mean = mean(cited_by_count),
    mncs = cited_by_count / pmax(field_mean, 1)
  )

# Fractional counting
works ▷
  unnest(authorships, names_sep = "_") ▷
  group_by(id) ▷
  mutate(frac = 1 / n()) ▷
  ungroup()
```

## D.5 Companion package functions

The `scientometricsInR` package provides helper functions used throughout the book:

| Function | Purpose | Example |
|---|---|---|
| `fetch_openalex(...)` | Cached, rate-limited OpenAlex queries | `fetch_openalex(entity = "works", ...)` |
| `dedupe_by_doi(df)` | Remove duplicate records by DOI | `works ▷ dedupe_by_doi()` |
| `compute_h_index(citations)` | Compute h-index from citation vector | `compute_h_index(c(10, 8, 5, 3, 1))` → 3 |
| `field_normalize(cites, means)` | MNCS normalisation (vector) | `field_normalize(c(20, 5), c(10, 10))` → `c(2.0, 0.5)` |
| `compute_mncs(df)` | MNCS for a grouped data frame | `df ▷ compute_mncs()` |
| `build_coauth_graph(works)` | Co-authorship igraph from works | `build_coauth_graph(works)` |
| `kleinberg_bursts(kw, dates)` | Keyword burst detection | `kleinberg_bursts(keywords, dates)` |
| `palette_sci(n)` | Viridis colour palette | `palette_sci(4)` |

| Function | Purpose | Example |
|---|---|---|
| `theme_sci()` | Minimal ggplot2 theme | `+ theme_sci()` |

## D.6 Useful keyboard shortcuts

| Action | RStudio | VS Code |
|---|---|---|
| Run current line/selection | Ctrl+Enter | Ctrl+Enter |
| Run current chunk | Ctrl+Shift+Enter | Ctrl+Shift+Enter |
| Insert pipe `\|>` | Ctrl+Shift+M | — |
| Insert assignment `<-` | Alt+- | — |
| Render document | Ctrl+Shift+K | — |
| Go to file | Ctrl+. | Ctrl+P |